Clinical Biochemistry:

1. What is the normal blood pH?
    a. 7.35–7.45
    b. 5.55–6.45
    c. 6.35–7.35
    d. 7.45–7.55
2. Which of the following represents the usual reference range for arterial PCO_2?
    a. 35–45 mmHg
    b. 45–55 mmHg
    c. 25–35 mmHg
    d. 25–35 mmHg
3. Which of the following is the usual reference range for plasma bicarbonate (HCO_3^-)?
    a. 22–26 mmol/L
    b. 30–36 mmol/L
    c. 10–16 mmol/L
    d. 18–22 mmol/L
4. Which statement best defines metabolic acidosis?
    a. A primary decrease in plasma bicarbonate concentration
    b. A primary increase in arterial PCO_2
    c. A primary increase in plasma bicarbonate concentration
    d. A primary decrease in arterial PCO_2
5. Which laboratory change characterizes metabolic alkalosis?
    a. A primary increase in HCO_3^-
    b. A primary decrease in HCO_3^-
    c. A primary increase in PCO_2
    d. A primary decrease in blood pH caused by carbon dioxide retention
6. A patient develops carbon dioxide retention following respiratory depression. Which primary disorder is most likely?
    a. Respiratory acidosis
    b. Metabolic acidosis
    c. Metabolic alkalosis
    d. Respiratory alkalosis
7. Which change defines respiratory alkalosis?
    a. A primary decrease in arterial PCO_2
    b. A primary increase in arterial PCO_2
    c. A primary decrease in plasma bicarbonate due to gastrointestinal loss
    d. A primary increase in plasma bicarbonate due to acid loss
8. A patient with metabolic acidosis begins breathing deeply and rapidly. Which mechanism explains this compensatory response?
    a. The lungs increase carbon dioxide elimination, lowering arterial PCO_2
    b. The kidneys reduce hydrogen ion excretion to retain acid
    c. The lungs retain carbon dioxide to restore bicarbonate
    d. The kidneys immediately eliminate all circulating bicarbonate
9. A patient develops metabolic alkalosis after repeated vomiting. Which response is most consistent with respiratory compensation?
    a. Decreased ventilation to retain carbon dioxide and increase PCO_2

b. Increased ventilation to lower PCO_2
c. Increased renal bicarbonate retention in every circumstance
d. Increased hydrogen ion production by the lungs

10. A patient with acute respiratory acidosis has an elevated arterial PCO_2. Which mechanism provides the main compensation as the disorder becomes chronic?
    a. The kidneys increase hydrogen ion excretion and conserve or generate bicarbonate.
    b. The lungs increase bicarbonate production directly.
    c. The kidneys excrete more bicarbonate to lower blood pH.
    d. The respiratory system immediately eliminates the entire acid load.
11. A patient develops respiratory alkalosis due to sustained hyperventilation. Which renal response helps compensate for the disturbance?
    a. Increased bicarbonate excretion, reducing plasma bicarbonate concentration
    b. Increased bicarbonate reabsorption to raise blood pH further
    c. Reduced bicarbonate excretion and increased hydrogen ion retention
    d. Complete cessation of renal acid–base regulation
12. A patient's arterial blood pH is 7.28. Which interpretation is most appropriate?
    a. Acidemia, indicating an increased hydrogen ion concentration
    b. Alkalemia caused by a low hydrogen ion concentration
    c. Normal acid–base status because the pH is close to 7.35
    d. Respiratory alkalosis because hydrogen ions are elevated
13. A patient with impaired ventilation develops carbon dioxide retention. Which change is expected if other factors remain constant?
    a. Increased hydrogen ion concentration and decreased pH
    b. Increased pH because carbon dioxide is alkaline
    c. Decreased hydrogen ion concentration
    d. Increased bicarbonate loss as the immediate primary event
14. Why can the bicarbonate buffer system continue to help regulate blood pH more effectively than a closed buffer system?
    a. It converts carbonic acid into a gas that can be exhaled.
    b. It permanently destroys all hydrogen ions.
    c. It removes all bicarbonate through the kidneys immediately.
    d. It prevents carbon dioxide from entering the blood.
15. A patient has increased acid production following intense tissue hypoxia. Which physiological response provides the most immediate compensation for metabolic acidosis?
    a. Increased ventilation, reducing arterial PCO_2
    b. Increased renal bicarbonate generation over several days
    c. Reduced ventilation to conserve carbon dioxide
    d. Increased gastric acid secretion
16. Which statement best describes the role of the kidneys in acid–base regulation?
    a. They reabsorb filtered bicarbonate and generate new bicarbonate while excreting acid.
    b. They regulate pH exclusively by exhaling carbon dioxide.
    c. They temporarily bind hydrogen ions but never excrete them.
    d. They compensate for every acid–base disorder within seconds.

17. A patient with diabetic ketoacidosis has pH 7.21 (low), HCO_3^- 11 mmol/L (low), and PCO_2 25 mmHg (low). Which primary disorder best explains these results?
    a. Metabolic acidosis
    b. Metabolic alkalosis
    c. Respiratory acidosis
    d. Respiratory alkalosis
18. A patient has persistent vomiting. Results show pH 7.51 (elevated), HCO_3^- 34 mmol/L (elevated), and PCO_2 46 mmHg (slightly elevated). What is the most likely primary disturbance?
    a. Metabolic alkalosis
    b. Respiratory acidosis
    c. Respiratory alkalosis
    d. Metabolic acidosis
19. A patient receives excessive respiratory-depressant medication and develops pH 7.25 (low), PCO_2 62 mmHg (elevated), and HCO_3^- 26 mmol/L (normal). Which disorder is most likely?
    a. Acute respiratory acidosis
    b. Chronic respiratory alkalosis
    c. Metabolic acidosis
    d. Metabolic alkalosis
20. A patient with marked anxiety is breathing rapidly. Arterial Blood Gas (ABG) shows pH 7.53 (elevated), PCO_2 27 mmHg (low), and HCO_3^- 22 mmol/L (normal). Which interpretation is best?
    a. Primary respiratory alkalosis
    b. Primary metabolic alkalosis
    c. Primary respiratory acidosis
    d. Primary metabolic acidosis
21. A patient has metabolic acidosis. Which change is expected as an appropriate respiratory compensatory response?
    a. A decrease in PCO_2
    b. An increase in PCO_2
    c. A substantial increase in bicarbonate as the immediate respiratory response
    d. No change in ventilation
22. A patient with chronic obstructive pulmonary disease has chronic carbon dioxide retention. Why might their bicarbonate concentration be elevated?
    a. The kidneys retain and generate bicarbonate during chronic respiratory acidosis.
    b. The lungs directly synthesize bicarbonate to compensate.
    c. Carbon dioxide retention always causes bicarbonate loss.
    d. Elevated bicarbonate proves the patient has metabolic alkalosis as the sole disorder.
23. Which statement about compensation is most accurate?
    a. Respiratory compensation is generally faster than renal compensation.
    b. Compensation always returns the underlying process to normal.
    c. Renal compensation is completed within seconds.
    d. Compensation necessarily causes the opposite acid–base disorder.
24. A patient with severe diarrhea has pH 7.30 (low) and decreased bicarbonate. Which mechanism most directly explains the metabolic component?

a. Loss of bicarbonate-rich intestinal fluid
b. Excessive loss of gastric hydrochloric acid
c. Primary retention of carbon dioxide
d. Excessive carbon dioxide elimination as the initial cause

25. A patient with suspected acute viral hepatitis has ALT 1,250 U/L (elevated) and AST 980 U/L (elevated). Which interpretation best explains the purpose of measuring these enzymes?
    a. They provide evidence of hepatocellular injury.
    b. They directly measure the liver's capacity to synthesize clotting factors.
    c. They specifically measure bile flow through the common bile duct.
    d. They directly quantify hepatic glycogen storage.
26. A patient has itching, dark urine, pale stools, increased ALP, and increased GGT. Which process should be investigated first?
    a. Cholestasis or impaired bile flow
    b. Isolated reduction in albumin synthesis
    c. Increased muscle protein breakdown alone
    d. Reduced production of unconjugated bilirubin
27. A patient with suspected acute liver failure has a prolonged PT/INR. Why is this result clinically important?
    a. It may indicate impaired synthesis of coagulation factors and can change relatively quickly.
    b. It indicates increased production of clotting factors.
    c. It specifically identifies viral hepatitis.
    d. It measures the degree of bile pigment formation.
28. A patient with chronic liver disease has a low serum albumin concentration. What is the most appropriate interpretation?
    a. It may indicate reduced hepatic synthesis, but other causes must be considered.
    b. It confirms acute hepatocyte necrosis.
    c. It proves extrahepatic biliary obstruction.
    d. It directly measures the rate of bilirubin conjugation.
29. A patient has increased unconjugated bilirubin, normal ALT and ALP, and evidence of increased red-cell destruction. Which mechanism best explains the bilirubin elevation?
    a. Excessive bilirubin production from heme breakdown exceeds the liver's ability to process it.
    b. Complete obstruction of the common bile duct
    c. Selective failure of albumin synthesis
    d. Reduced production of bilirubin from heme
30. A patient with obstructive jaundice has increased conjugated bilirubin. Which property explains why conjugated bilirubin may appear in urine?
    a. It is water-soluble and can be excreted through the kidneys.
    b. It is highly fat-soluble and cannot circulate in blood.
    c. It is converted directly into albumin.
    d. It is identical to stercobilin.
31. A patient with extrahepatic biliary obstruction develops pale stools. Which explanation is most appropriate?
    a. Less bilirubin reaches the intestine, reducing formation of stool pigments.
    b. Excess stercobilin is being formed in the intestine.

c. Increased albumin synthesis bleaches the stool.
d. Unconjugated bilirubin is directly excreted in the stool without processing.

32. A patient has increased ALP but normal GGT and no bilirubin elevation. Which additional consideration is particularly important before attributing the abnormality to the liver?
    a. Bone disease or another non-hepatic source of ALP
    b. Acute hepatic synthetic failure as the only possible explanation
    c. Complete biliary obstruction as a confirmed diagnosis
    d. Increased unconjugated bilirubin production as the sole explanation
33. A patient with suspected drug-induced liver injury has a large increase in ALT and AST shortly after a medication overdose. Bilirubin rises later. Which interpretation best fits this sequence?
    a. Early hepatocellular injury followed by impaired bilirubin handling
    b. An isolated primary defect in albumin synthesis
    c. Early cholestasis with no hepatocellular involvement
    d. Normal physiological adaptation to medication
34. A patient with suspected hepatocellular carcinoma has an elevated alpha-fetoprotein (AFP). What is the most appropriate interpretation?
    a. AFP can support evaluation in the appropriate context, but it is not sufficiently specific to diagnose cancer by itself.
    b. AFP elevation alone definitively establishes the diagnosis.
    c. AFP directly measures the rate of bile secretion.
    d. A normal AFP excludes hepatocellular carcinoma.
35. A patient has an increased ALP concentration, but it is unclear whether the source is the liver or bone. Which additional test is particularly useful?
    a. GGT
    b. Amylase
    c. CK-MM
    d. PSA
36. A patient has an elevated LDH concentration during an investigation for suspected liver disease. Which conclusion is most appropriate?
    a. LDH is nonspecific, so hemolysis, malignancy, and other tissue injuries must also be considered.
    b. LDH elevation proves that liver injury is the only abnormality.
    c. LDH is produced exclusively by hepatocytes.
    d. LDH elevation confirms a pancreatic disorder.
37. A patient presents with severe epigastric pain radiating to the back, raising suspicion of acute pancreatitis. Both amylase and lipase are available. Which statement best supports the selection of lipase?
    a. Lipase is more specific for pancreatic injury and remains elevated longer than amylase.
    b. Lipase is primarily a marker of bone formation.
    c. Amylase is found exclusively in the pancreas, whereas lipase is found in the salivary glands.
    d. Lipase is a specific marker of myocardial injury.
38. A patient develops severe muscle pain and dark urine after strenuous exercise. Laboratory results show a markedly increased CK-MM concentration. Which condition is most consistent with this result?

a. Rhabdomyolysis
b. Cholestasis
c. Acute pancreatitis
d. Prostate enlargement

39. A patient has elevated AST after intense physical exercise. ALT is not substantially increased, and CK-MM is markedly elevated. Which explanation is most likely?
    a. The pattern supports skeletal muscle injury as a possible source of the elevated AST.
    b. AST elevation must originate exclusively from the liver.
    c. The result proves that the patient has biliary obstruction.
    d. The findings confirm acute pancreatitis.
40. A patient is being investigated for a disorder involving increased bone breakdown. Which enzyme is more specific for osteoclast activity?
    a. TRAP
    b. ALT
    c. CK-MB
    d. Lipase

Clinical Biology:

1. A 4-year-old child presents with recurrent episodes of itchy, dry, inflamed skin. The symptoms improve with treatment but return periodically. Which condition best explains?
    a. Atopic dermatitis
    b. Acute bacterial skin infection
    c. A single episode of allergic contact dermatitis
    d. A non-inflammatory pigmentary disorder
2. Which combination represents the classic atopic triad?
    a. Eczema, asthma, and allergic rhinitis
    b. Psoriasis, asthma, and urticaria
    c. Eczema, diabetes mellitus, and allergic rhinitis
    d. Asthma, bacterial infection, and psoriasis
3. Which finding is most characteristic of adolescent or adult Atopic Dermatitis?
    a. Thick, dry, lichenified plaques with prominent skin markings
    b. Exclusively wet lesions on the cheeks
    c. A complete absence of itching
    d. Lesions restricted to the nappy area
4. A patient with AD has a mutation affecting the filaggrin gene. Which consequence is most directly associated with this abnormality?
    a. Impaired skin barrier function and reduced natural moisturizing factor
    b. Increased production of natural moisturizing factor
    c. Improved protection against environmental allergens
    d. Complete suppression of inflammatory cytokines
5. Why does a defective epidermal barrier contribute to severe dryness in Atopic Dermatitis?
    a. It increases transepidermal water loss.
    b. It prevents all water movement through the epidermis.
    c. It increases the production of skin lipids in every patient.
    d. It blocks the entry of environmental irritants.

6. A patient with AD experiences severe itching that triggers repeated scratching. Which cytokine is identified as a major initiator of pruritus?
    a. IL-31
    b. Insulin
    c. Erythropoietin
    d. Albumin
7. Which sequence correctly describes the itch–scratch cycle in Atopic Dermatitis?
    a. IL-31-mediated itch → scratching → keratinocyte damage → increased alarmin release → more inflammation and itch
    b. Scratching → stronger barrier → reduced inflammation → increased itching
    c. Reduced water loss → scratching → complete healing → recurrent dryness
    d. Moisturization → keratinocyte injury → increased infection in every patient
8. Which combination contains only environmental or external triggers?
    a. Dust mites, sweat, dry winter air, and Staphylococcus aureus
    b. Filaggrin mutation, inherited gene variation, and genetic predisposition
    c. Reduced NMF, increased TEWL, and defective filaggrin function
    d. Lichenification, excoriation, and fissure formation
9. A patient has thickened, leathery skin with exaggerated skin lines following months of repeated scratching. What is the correct term for this secondary lesion?
    a. Lichenification
    b. Vesicle
    c. Excoriation
    d. Erythema
10. Which approach is most appropriate for diagnosing Atopic Dermatitis?
    a. Combine the patient's history with a physical examination and clinical criteria
    b. Rely exclusively on a serum IgE concentration
    c. Use a skin biopsy as the only definitive test
    d. Diagnose the condition from a single liver enzyme measurement